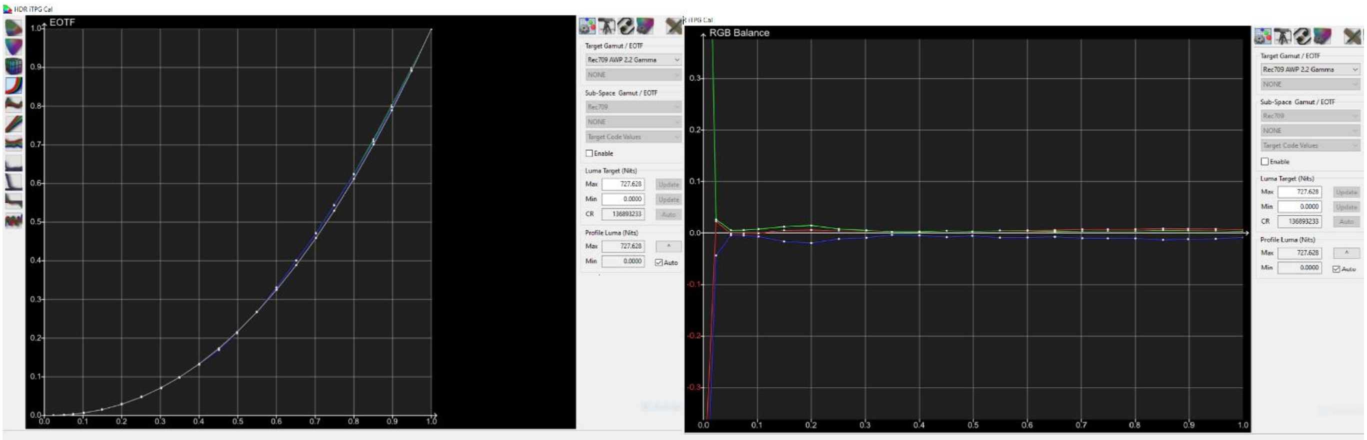
HDR Manual Calibration with Tone Curve Upload

The following guide will provide details on how to manually calibrate HDR on a 2019 LG OLED using the internal Test Pattern Generator (iTPG), the 2 Point White Balance controls, and upload of your specific panel peak luminance and, if desired, your own custom Tone Mapping curves via Device Control Template for LG.

Unlike Dolby Vision which requires a full 22 Point White Balance adjustment, best results for HDR will be obtained by using just the 2 Point White Balance controls only. LG's 22 Point controls for HDR are based on 10-bit Code Values, the majority of which are all tightly bunched together within the mid luminance range. Incorrect adjustment of these points will cause image artefacts such as severe banding.

In addition to this, with the TV in HDR mode but in gamma space, it already tracks 2.2 Gamma very well natively, and has relatively good RGB Balance throughout the luminance range. What this means in real terms is that if you have White Balance accurately pre-calibrated to your desired white point in the Service Menu, you already have a very solid base-line for HDR to work from as can be seen with this HDR pre-calibration grey scale sweep



Minor tweaks of just the 2 points will give great results for when the TV's internal processing converts from 2.2 Gamma to PQ EOTF.

Furthermore, the factory 3x3 colour management matrix is already very good, so there is no need for any "Matrix LUT" upload. User Menu CMS controls must not be used, as HDR gamut mapping is based from the factory default values.

To begin, you must ensure you have the latest version of Device Control (currently 2.0.19) and the correct LG OLED Template for Device Control for your TV model. At the time of writing this guide, this will be LG 2019 Template Rev.5.

For the upload of Peak Luminance and Custom Tone Curves, you will also require the LG 2019 PQ Curve Template, 2019 PQ Rev.2

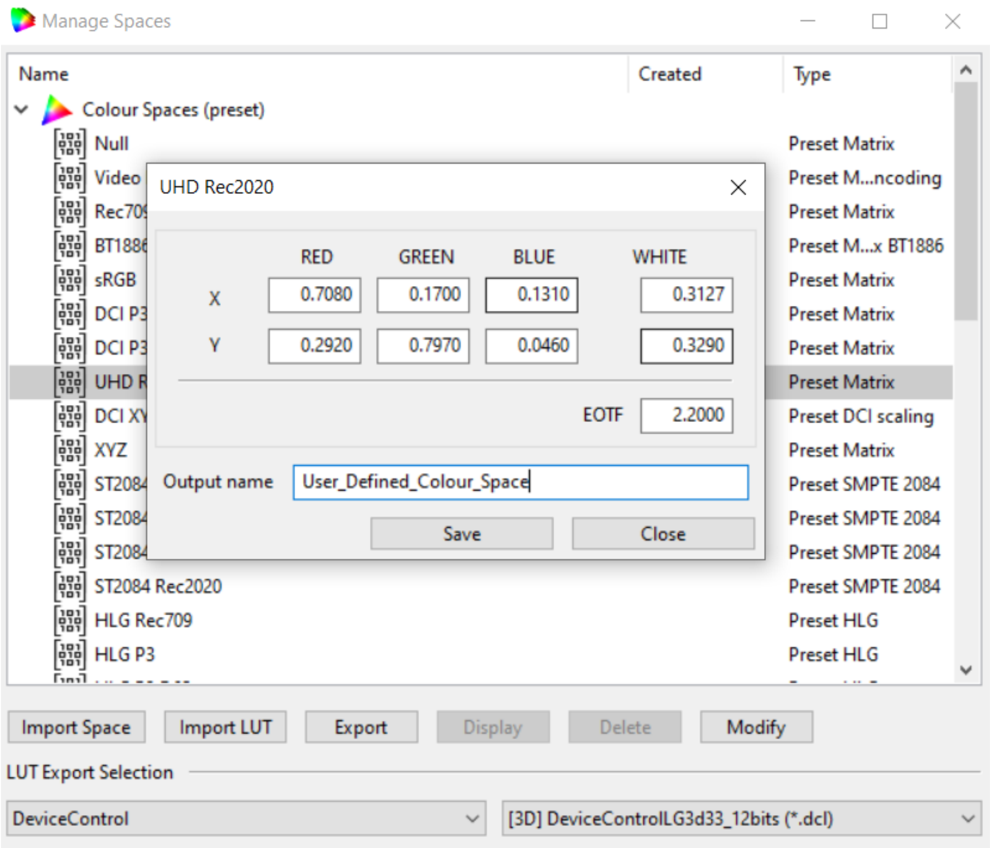
It is also advised to make your adjustments starting from a pre-calibrated Service Menu White Balance colour temperature. This can be the same pre calibrated Service Menu White Balance (SM WB) used for SDR 3D LUT calibration. There is no need to pre-calibrate separate SM WB settings just for Dolby Vision or HDR.

I personally use an AWP (Alternate White Point) for all my calibrations on LG OLED. It was obtained by perceptually matching to a Pioneer Kuro Plasma calibrated to true D65 with a 5nm Jeti 1501 Spectroradiometer and some images within this guide make reference to this AWP. However, the AWP I use is only applicable for users that have a probe matched to a reference 5nm Spectroradiometer and will not produce the same results when using an unmatched colorimeter, or a colorimeter using either Generic CMF or the WRGB OLED EDR.

(Note: The use of an alternate white point is a personal choice and is not required. Targeting an alternate white point should only be done if you have a Spectro or a colorimeter that has been “profiled” to one)

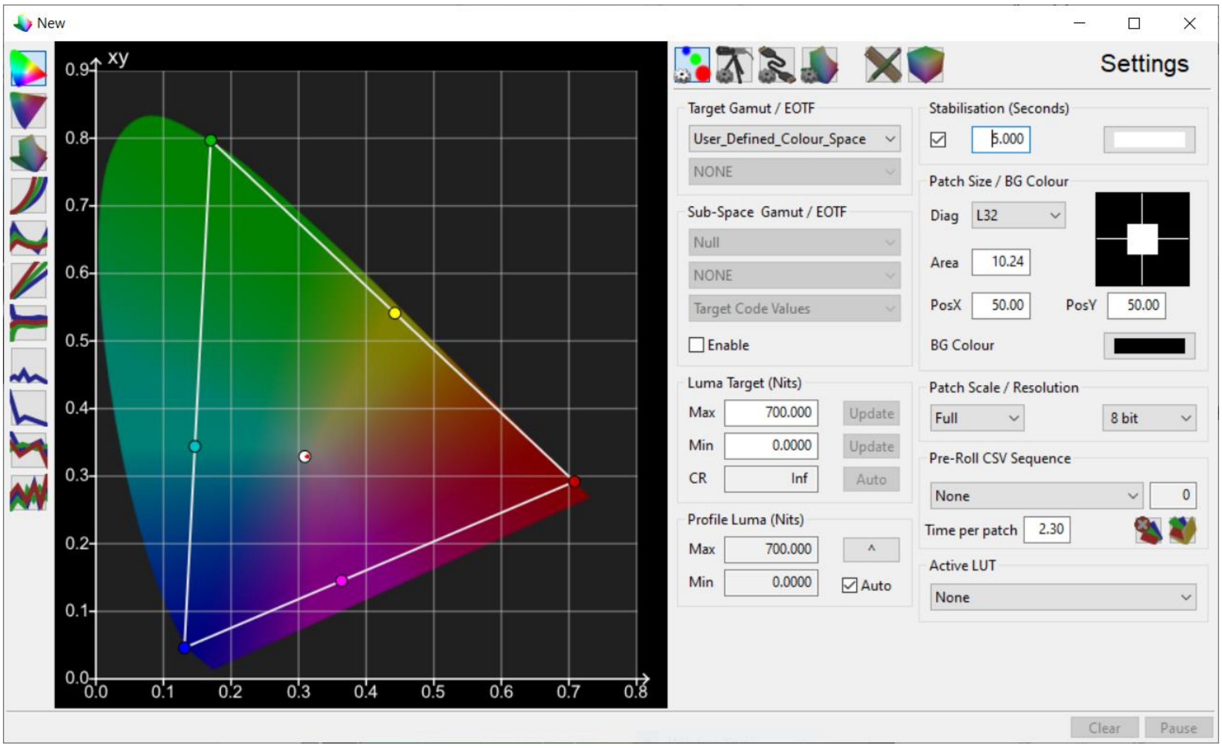
When calibrating HDR with this method, you will only be adjusting 2 Points of the white balance controls, but as with the Dolby Vision calibration you will want to use a Target Gamut based on a Power Law Gamma of 2.2. The conversion to PQ EOTF is done automatically based from these underlying adjustments. So, with your white point decided and your SM WB pre-calibrated, ensure you have ColourSpace set up to calibrate to your desired white point and 2.2 Gamma. Some of the images in this guide will show my target AWP settings, but users will have to select their own settings based to their desired white point target. For example, users wishing to target D65 can use the built in UHD Rec.2020 Target Gamut, adjusting the EOTF to 2.2 Gamma (modified Target Gamut included with these guides). Users wishing to use their own custom AWP will need to modify a built in Target Gamut to create one with their own AWP, ensuring to set the gamma target of 2.2.

To do this from ColourSpace, open Manage Spaces



Select UHD Rec2020 and click Modify to edit the desired White Point coordinates , ensuring to also edit the EOTF value to 2.2000. Give the new modified Colour Space a name and click Save.

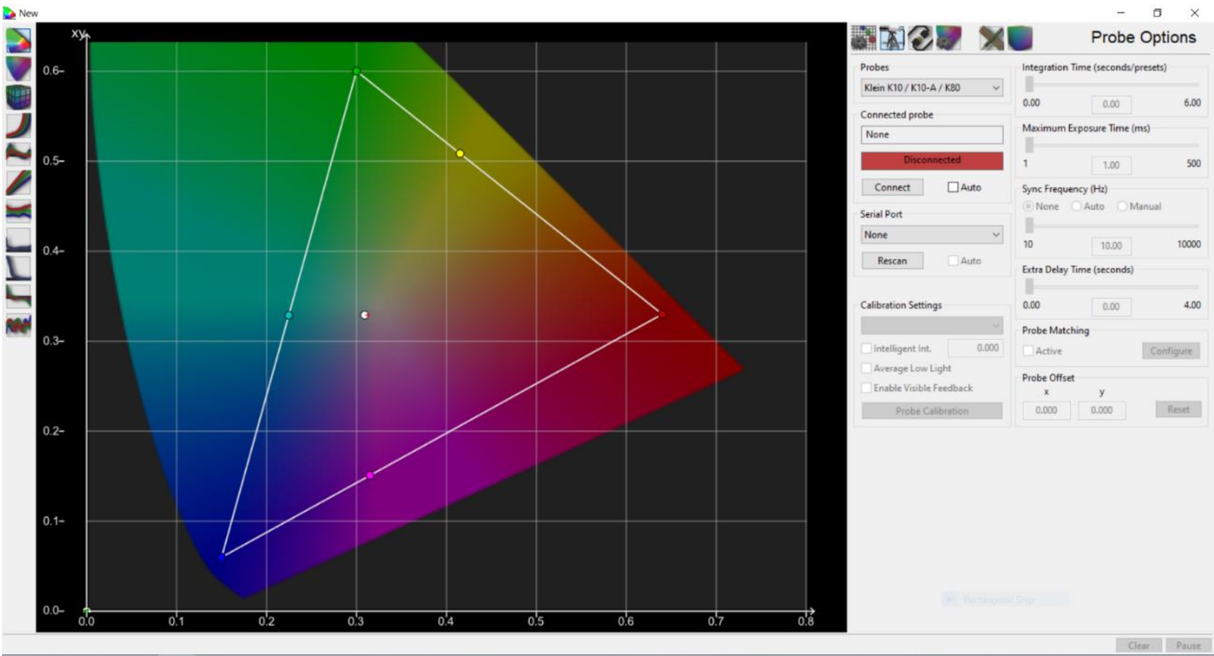
Open a ColourSpace Profile window and set this Target Gamut now.



While in this screen and after setting Target Gamut, set Stabilisation to 5 seconds and tick to enable. Set Patch Size Diag to L32. Place a tick in Profile Luma (Nits) Auto and set Patch Scale to 0-255

Change to the Probe Options tab and connect your colorimeter, remembering to configure Probe Matching if required, or selecting a meter pre-set (e.g. EDR) if your meter is not profiled to a Spectro. Set all other probe options as you normally would (e.g. Sync Mode AIO etc, for i1 Display Pro 3 users)

Set a minimum Extra Delay Time (seconds) of 0.50 to minimise the risk of any sync read issues with the iTPG.

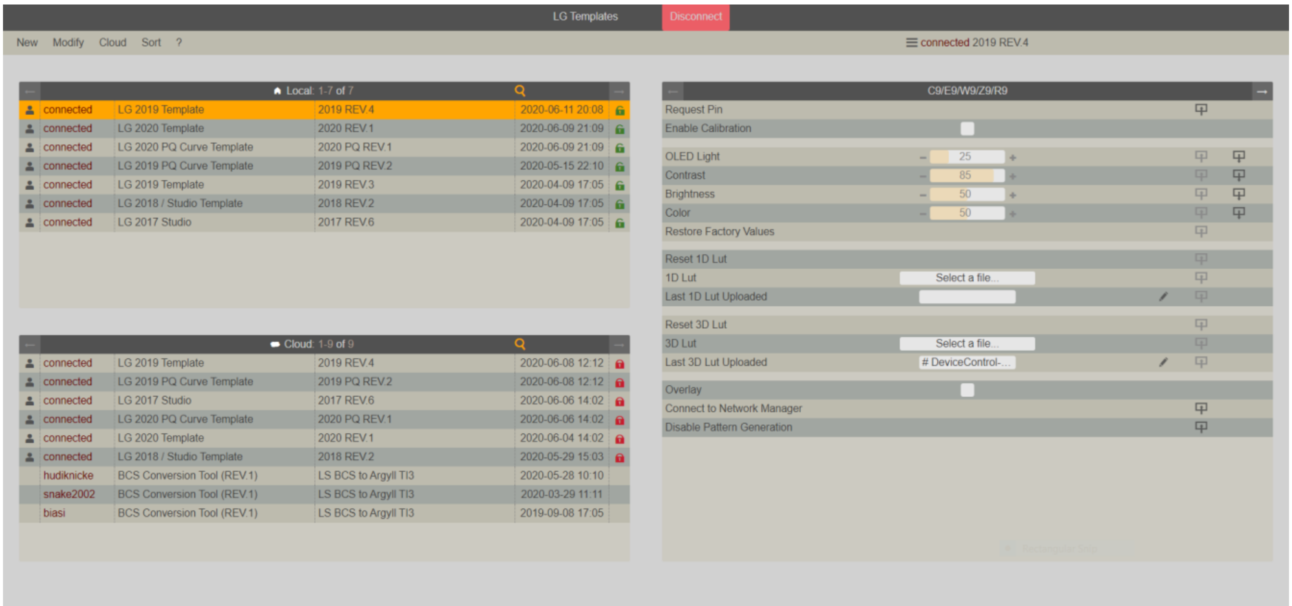


Change to the Hardware Options tab in ColourSpace in preparation to connect to the iTPG

Launch Device Control, open your web browser and navigate to http://127.0.0.1

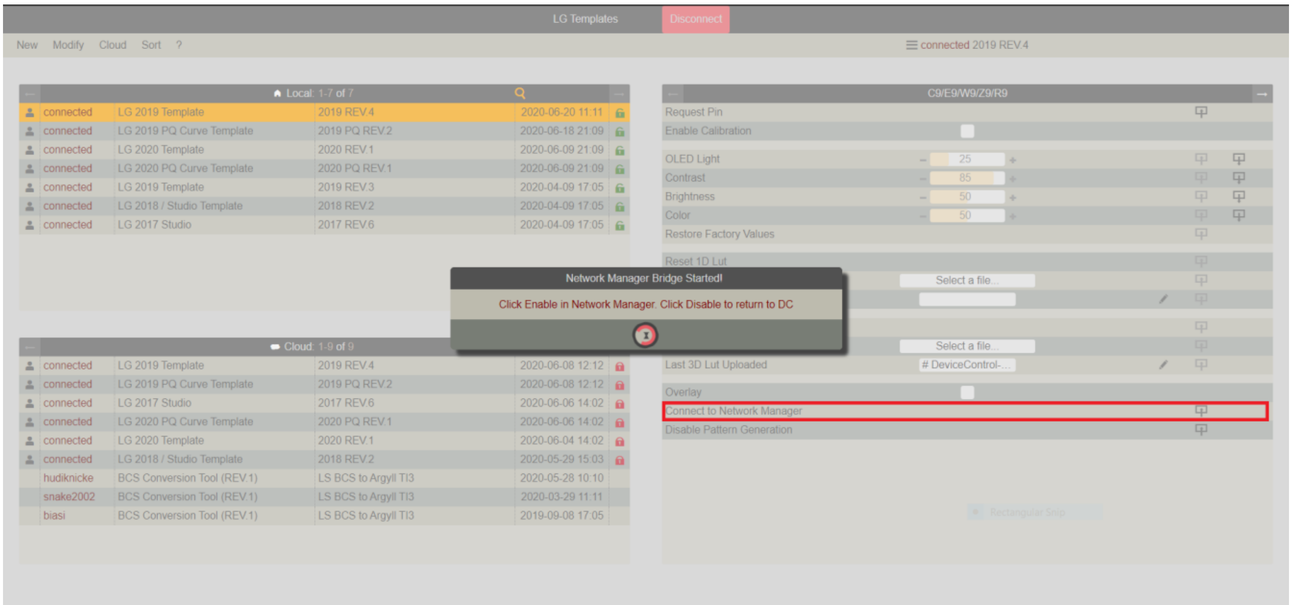
Select the LG 2019 Template Rev.4, and click the green Connect button.

Connect to your TV by Requesting PIN, then entering and sending the PIN displayed on screen.

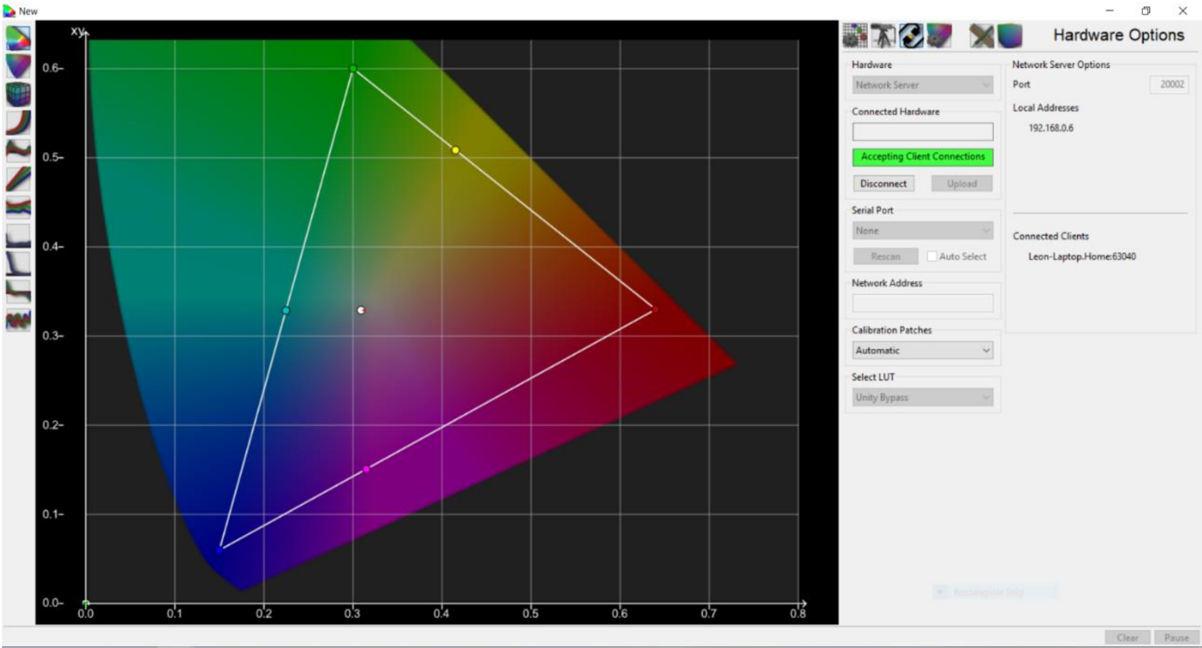


If you have an HDR blank video file to put onto a USB drive, insert this into the TV USB port and play it via the internal media player now. If you do not have an HDR blank video file, you can play some HDR content e.g. via a streaming service, but ensure it is sufficiently long enough to last the duration of the calibration (e.g. movie length content). Failure to do so could mean the iTPG dropping back out of HDR mode and you will end up adjusting the SDR luminance levels.

With the TV now in HDR mode, select “Connect to Network Manager” in Device Control Template, and switch (e.g. Alt+Tab) back to ColourSpace

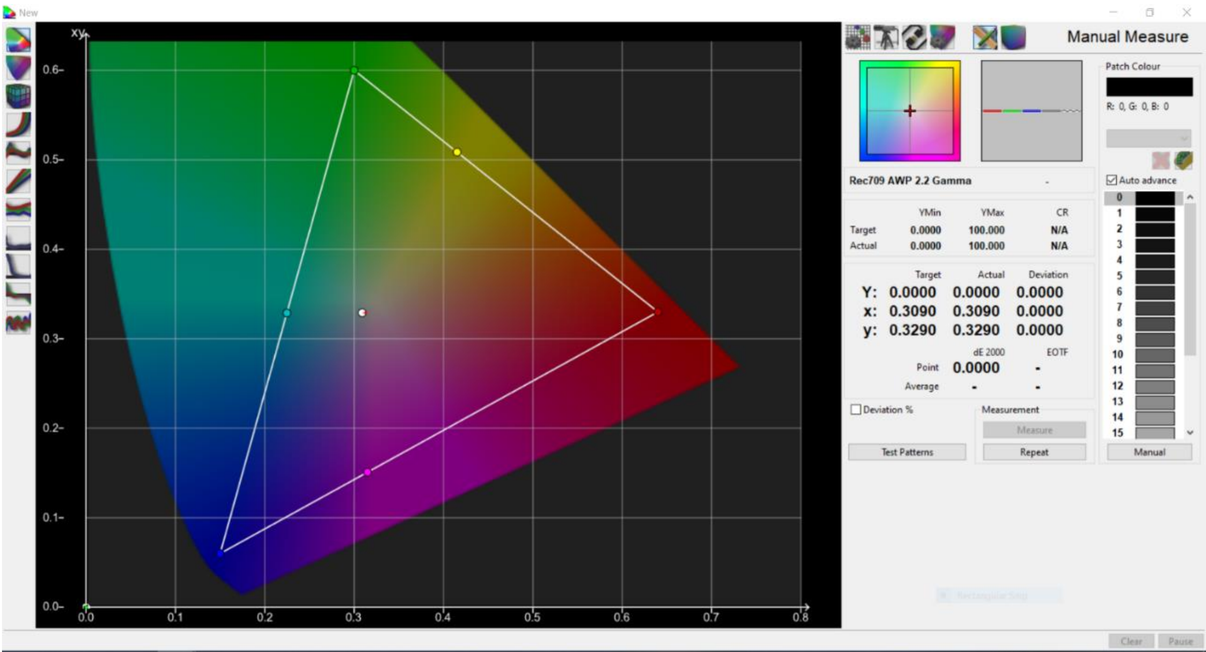


Select Network Server from the Hardware list and click Connect. You will see your computer name shown in the Connected Clients. Once it appears, select Automatic under Calibration Patches. You are now connected to the iTPG in HDR mode and have the ability to display the required patches needed for this calibration.



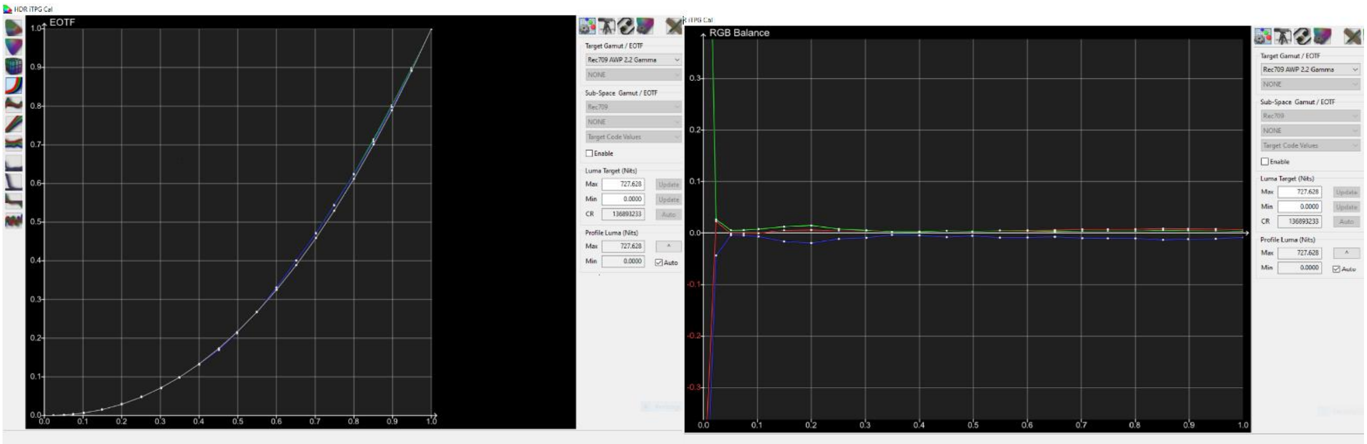
Make sure you are in either HDR Cinema (not Cinema Home) or HDR Technicolor at this point, and that your Colour Temperature is set to the same one you have pre-calibrated in the Service Menu. Please also ensure all extra video processing is disabled, such as Dynamic Tone Mapping, Noise Reduction, etc.

Change to the Manual Measure tab, click “Presets” and load the provided LG OLED 22 Point White Balance – Full Range.csv patch sequence



Place a tick in “Auto Advance” and click Measure to take a pre-calibration grey scale sweep. This is purely to give an idea of how well your display is tracking 2.2 Gamma and how good the RGB Balance is. Do not attempt to adjust any of the User Menu 22 Point White Balance controls as the patches in this sequence do not align at all with any of them (with the obvious exception of Black and 100% White). It will also give you an idea as to whether you will eventually adjust both of the 2 Point controls, or just one of them. If you are measuring based from an accurately pre-calibrated Service Menu White Balance Colour Temperature, then you should end up with something similar to what was shown in the first image in this guide.

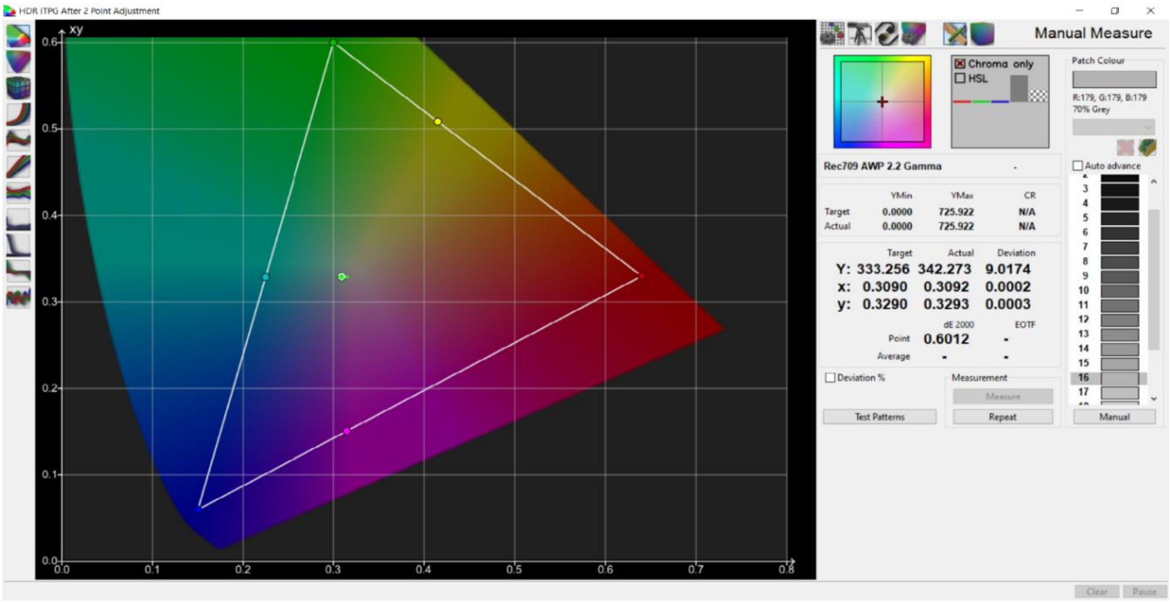
Here it is shown again for reference



Once this has completed, **you must remove the tick from "Auto Advance"** before progressing to the next step.

Examine the RGB Balance graph. As you can see above, RGB Balance is reasonable but could do with a "minor" tweak to both High and Low of the 2 Point controls. To do this, you will use the 20% and 70% Grey patches for Low and High respectively.

Select patch number 16 for 70% Grey, hover your mouse over the RGB bars to see an option of "Chroma Only" (as shown in the image below) and select it.

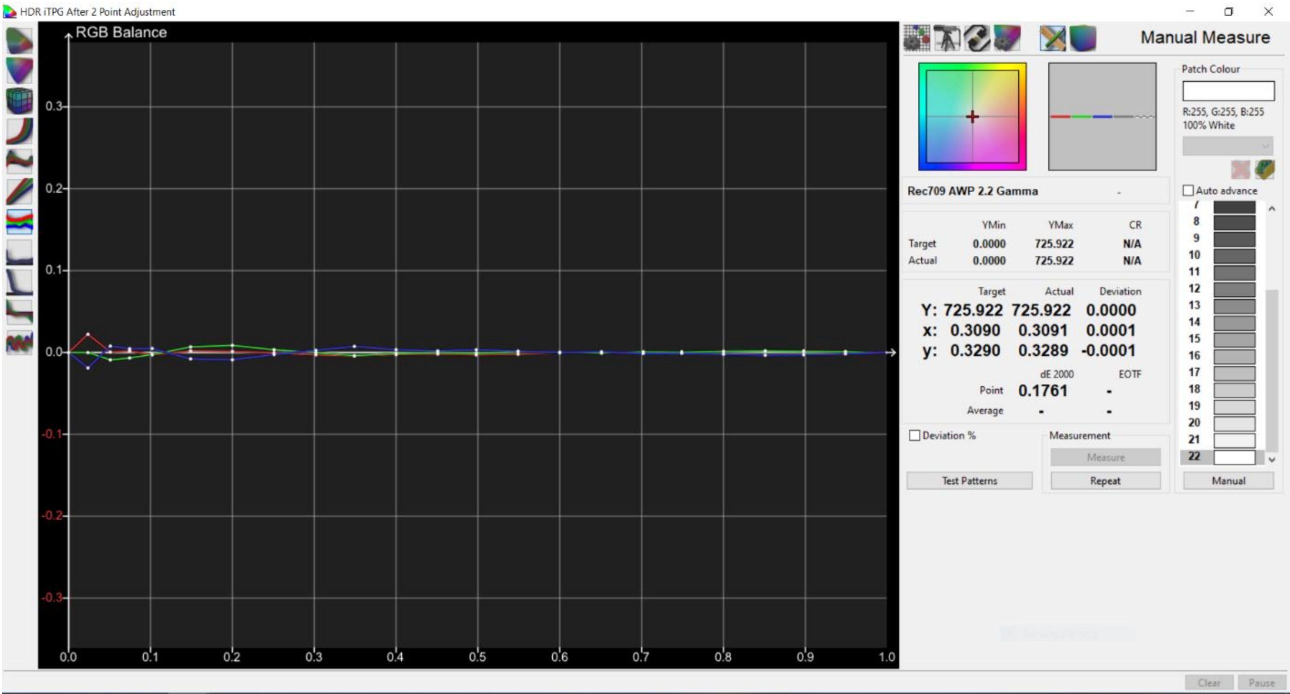


Click Repeat, then Measure and use the User Menu -> Picture Option -> Expert Controls -> White Balance -> 2 Points High to make adjustments to your target white point, **but do not attempt to correct for luminance.** The adjustments should be very minimal. Avoid adjusting the Green control if possible. In the example image shown above, all that was required for 2 Point High was Red -1, Green 0, Blue +1.

Select patch number 6 for 20% Grey and repeat the above for 2 Point Low. Once again, minimal adjustment with no attempt to correct for luminance is all that is required. For example, in this session, I adjusted 2 Point Low with Red 0, Green 0, Blue +1.

To verify the adjustments have balanced the full grey scale, place a tick back in Auto Advance and click measure.

As you can see here, those absolute minimal adjustments have done a very reasonable job off correcting the entire greyscale



Whilst is not perfect, for manual adjustments in HDR it is the absolute best you can do. You could get “perfect” grey scale in HDR with a Custom 1D LUT, but that has other issues, such as the much talked about “raised blacks”

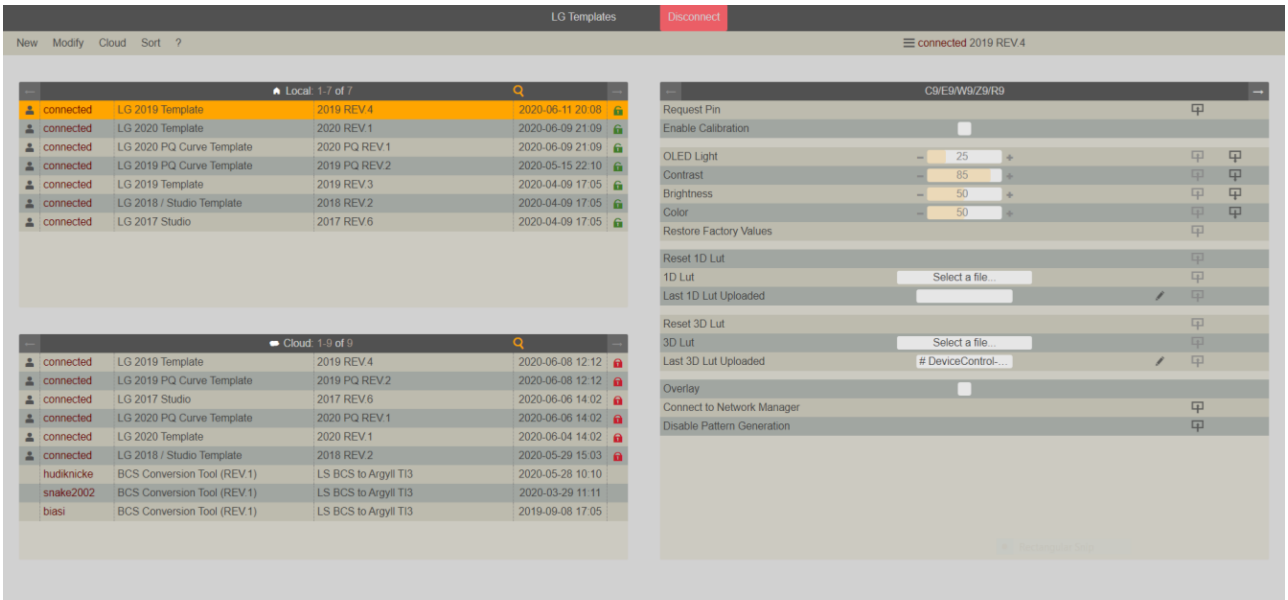
Tracking of 2.2 Gamma is still very reasonable too, meaning the conversion to PQ EOTF will work as expected. This will be shown later in the guide.

When you are happy with your adjustments, it is essential remember at this point to go back to the User Menu -> Expert Controls -> White Balance and select “Apply to All Inputs”

With measurements now complete, the iTPG can now be disconnected and disabled on the TV.

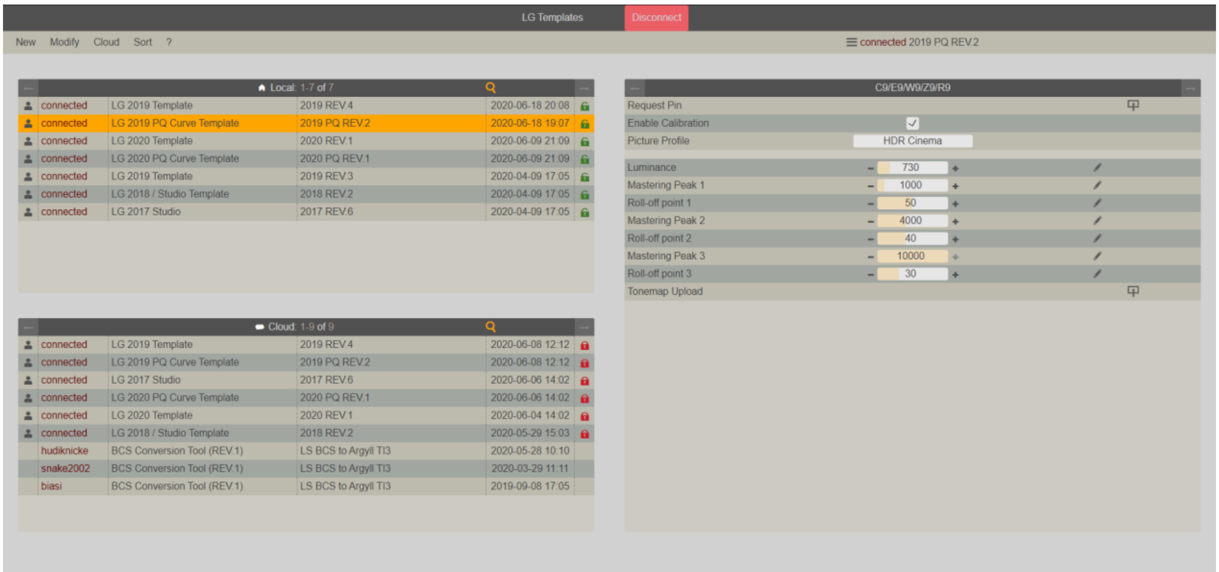
Change to the Hardware Options tab and click Disconnect.

Switch (Alt+Tab) to the Web browser window and click to **“Disable Pattern Generation”** to fully disable the iTPG.



Failure to do this will result in the iTPG continuing to be displayed despite not being able generate patterns from ColourSpace.

To upload your peak brightness for more accurate internal tone mapping, and Custom Tone Curves if desired, select the LG 2019 PQ Curve Template. There is no need to request/send a new PIN as you are still connected.

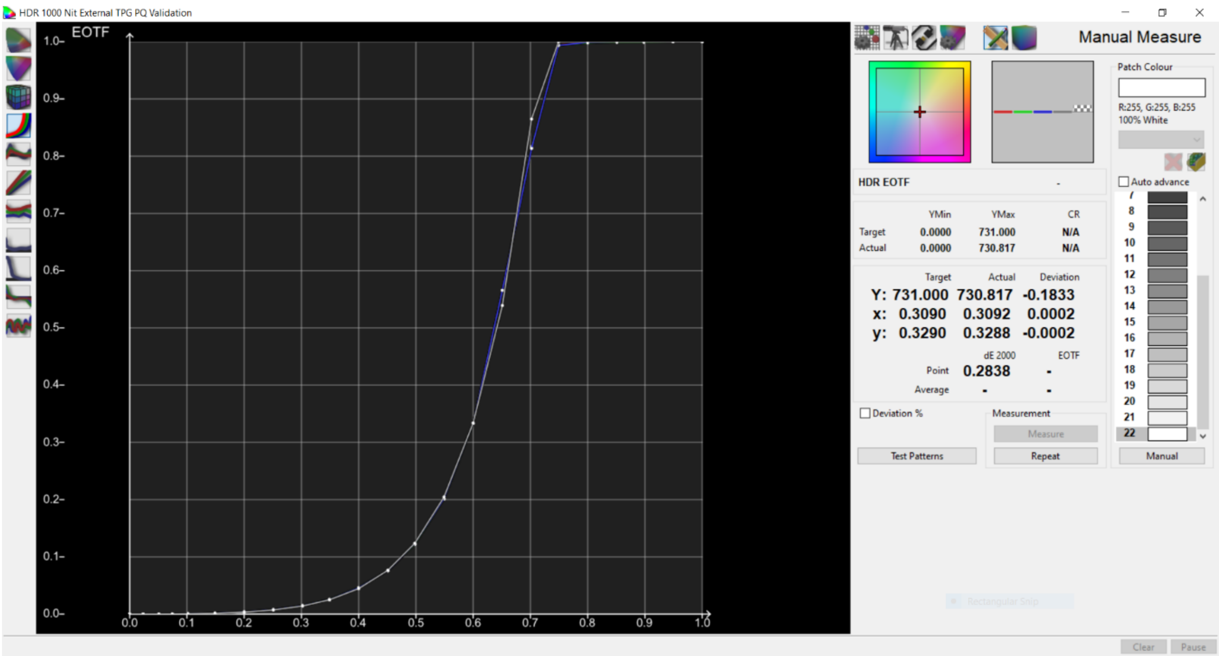


Tick "Enable Calibration" and make sure the picture mode you have adjusted is selected.

Using the +- symbols, set the peak luminance value measured at 100% White and click the pencil icon to set the value. You can leave the default tone curves or set your own custom values by setting them in the Roll-off point 1/2/3 sections, remembering to click the pencil icon each time to set the values. Once all values have been set, click Tonemap Upload, and then untick "Enable Calibration".

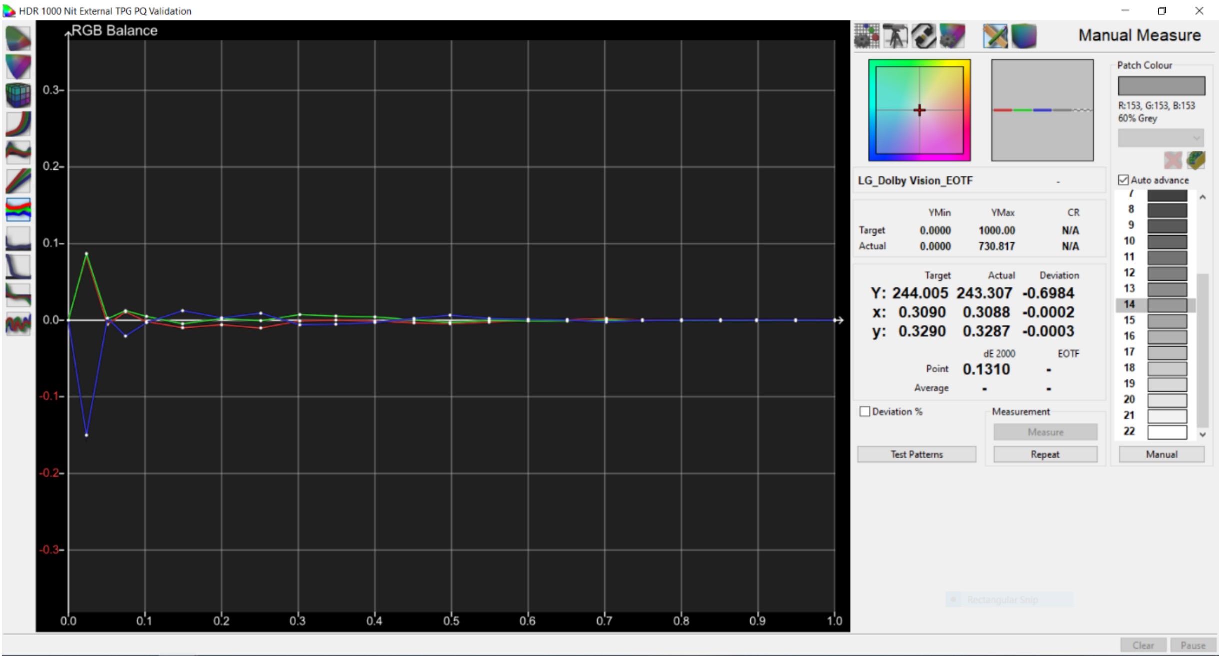
You can now disconnect Device Control Template and close the Device Control command window.

HDR calibration is complete. To see how the iTPG adjustments look when verified with an external device over HDMI, I have used an RPi PGenerator in combination with a HD Fury Integral 2 to send 1000 nit HDR metadata. The same 22 Point patch sequence has then been measured and can be seen below, but the results are the same even when sending 4000 and 10000 nit metadata.



For this example, the Custom Tone Curve upload was set to Roll-Off at 100% for all HDR metadata values. Note that this means the PQ EOTF should be tracked accurately until the panel peak luminance and then hard clip. The reason I set it this way is to see how well it performs after just 2 points being adjusted. Whilst it is not as "perfect" as the Dolby Vision calibration using 22 adjustment points, it is still tracking very, very well.

And the resulting RGB Balance throughout the grey scale is equally as good



Once again, considering only 2 points have been adjusted the results are very good, and there will be no raised blacks as seen with a Custom 1D LUT upload. If LG ever fix the Custom 1D LUT then HDR calibration can be revisited, but until that happens a simple 2 Point adjustment gives fantastic results.

Happy viewing,

Liberator72

Note: If you have calibrated HDR Cinema using the above method, you can copy all relevant settings across to HDR Technicolor without the need to remeasure. But you can, if you wish, upload a different set of custom tone curve values to each of these modes. For example, you can have HDR Cinema with the default curves, and experiment with various values in HDR Technicolor.

HDR Game can also be calibrated using this method, or you can apply all relevant settings from your calibrated mode to HDR game to achieve the same results. I personally do not set custom tone curves for HDR Game, instead choosing to use HGIG Mode within the User Menu Expert Controls to allow the games to handle the tone mapping themselves.